

A TOPOLOGICAL OBSTRUCTION TO THE LITERAL INVERSE ARBOREAL GALOIS PROBLEM

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. An inverse problem asks whether an arbitrary subgroup of the automorphism group of a regular rooted tree can occur as the image of an arboreal Galois representation. Taken literally, the answer is negative for a purely topological reason. The image of a continuous homomorphism from an absolute Galois group into the profinite automorphism group of a rooted tree is closed. On the other hand, the finitary automorphism group of the infinite rooted binary tree is a dense proper subgroup and hence is not closed. This gives a counterexample to the unrestricted arbitrary-subgroup formulation. It does not address the inverse problem for prescribed closed subgroups.

1. THE INVERSE PROBLEM AND ITS TOPOLOGY

Let T_d be the infinite regular rooted d -ary tree. An arboreal Galois representation associated with a polynomial and a basepoint is a continuous homomorphism

$$\rho : \text{Gal}(K^s/K) \longrightarrow \text{Aut}(T_d).$$

The absolute Galois group carries its Krull topology. The tree automorphism group carries the inverse-limit topology

$$\text{Aut}(T_d) \simeq \varprojlim_n \text{Aut}(T_{d,n}),$$

where $T_{d,n}$ is the finite truncation through level n .

The literal form of the inverse problem asks whether every abstract subgroup of $\text{Aut}(T_d)$ can arise as such an image. The following standard observation already imposes a necessary condition.

Proposition 1.1. *The image of every arboreal Galois representation is a closed subgroup of $\text{Aut}(T_d)$.*

Proof. The Krull-topological absolute Galois group is profinite and therefore compact. The group $\text{Aut}(T_d)$, being an inverse limit of finite discrete groups, is Hausdorff. The continuous image of a compact space is compact, and compact subsets of a Hausdorff space are closed. $\square$

2. THE FINITARY SUBGROUP

It remains to exhibit an abstract subgroup that violates the necessary closedness condition. We work with the binary tree

$$T = \{0, 1\}^*,$$

whose root is the empty word and whose children below a word w are $w0$ and $w1$.

For each $n \geq 0$, let T_n denote the truncation through level n . Every automorphism $h \in \text{Aut}(T_n)$ has a trivial extension to T : if a word of length greater than n is written uniquely as uv , with $|u| = n$, define

$$\iota_n(h)(uv) = h(u)v.$$

Define

$$F = \bigcup_{n \geq 0} \iota_n(\text{Aut}(T_n)).$$

Equivalently, F consists of those automorphisms whose local actions are trivial below some finite level. It is called the finitary automorphism group.

Lemma 2.1. *The subgroup F is dense in $\text{Aut}(T)$.*

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Proof. Let

$$\pi_m : \text{Aut}(T) \longrightarrow \text{Aut}(T_m)$$

be restriction. Given $a \in \text{Aut}(T_m)$, the trivial extension $\iota_m(a)$ lies in F and restricts to a . Hence

$$\pi_m(F) = \text{Aut}(T_m) \quad \text{for every } m.$$

Every nonempty basic open subset of the inverse-limit topology is determined by a finite-level restriction. Since F realizes every such restriction, every nonempty basic open set meets F . $\square$

Lemma 2.2. *The subgroup F is a proper subgroup of $\text{Aut}(T)$.*

Proof. Each group $\text{Aut}(T_n)$ is finite, so F is countable. The full automorphism group is uncountable. Indeed, the vertices

$$v_n = 1^n 0 \quad (n \geq 0)$$

are pairwise prefix-incomparable, and their descendant subtrees are disjoint. For each binary sequence $\epsilon = (\epsilon_n)_{n \geq 0}$, simultaneously interchange the two child subtrees below v_n precisely when $\epsilon_n = 1$. Disjointness of the supports makes this a tree automorphism, and distinct sequences give distinct automorphisms. Thus $\{0, 1\}^{\mathbb{N}}$ injects into $\text{Aut}(T)$. $\square$

Corollary 2.3. *The finitary subgroup F is not closed in $\text{Aut}(T)$.*

Proof. By Lemma 2.1, the closure of F is the full automorphism group. By Lemma 2.2, F is not the full automorphism group. $\square$

3. THE COUNTEREXAMPLE

Theorem 3.1. *There is a subgroup of the automorphism group of a regular rooted tree that cannot be the image of an arboreal Galois representation. In particular, the inverse problem with an unrestricted arbitrary-subgroup quantifier is false.*

Proof. Take the finitary subgroup

$$F \leq \text{Aut}(T)$$

of the infinite rooted binary tree. It is a subgroup by construction and is not closed by Corollary 2.3. If F were the image of an arboreal Galois representation, Proposition 1.1 would make it closed, a contradiction. $\square$

Remark 3.2 (Scope). The obstruction uses only the omission of a closedness hypothesis. It gives no nonrealizability result for any particular closed subgroup of $\text{Aut}(T_d)$. Thus the natural inverse problem for prescribed closed subgroups remains separate from the literal statement treated here.

4. DISCUSSION

The argument is independent of arithmetic properties of a polynomial. Its content is that the topology is part of the data of an arboreal image: an abstract subgroup that is not closed cannot occur, regardless of the field, basepoint, or dynamical construction. Consequently, a well-posed inverse problem must at least restrict the prescribed group to a closed subgroup of the profinite tree automorphism group.

APPENDIX A. CHECKLIST FOR VERIFICATION

- (1) Verify that the arboreal representation is continuous for the Krull and inverse-limit topologies.
- (2) Check compactness of the absolute Galois group and the Hausdorff property of $\text{Aut}(T_d)$.
- (3) Verify that every finite-level automorphism has the displayed trivial extension and hence that every finite projection of F is surjective.
- (4) Check that F is countable while $\text{Aut}(T)$ is uncountable.
- (5) Conclude that F is dense, proper, and nonclosed.
- (6) Keep the unrestricted abstract-subgroup statement separate from the closed-subgroup inverse problem.

REFERENCES

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